

DC Motor Parameter Estimation - Group 42

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Motivation and Goals

This project aims to merge mechatronics, control theory, and statistics to enhance interdisciplinary understanding. We will conduct physical measurements on a brushed DC motor and construct a steady-state model through statistical analysis. By modeling this motor, we aim to facilitate its utilization in future projects such as servo control, ball-on-beam regulation, and inverted pendulums.

System Description

The position of the motor's rotor is measured with a hall-effect absolute encoder (AS5600) while the current and voltage are measured with a shunt resistor current sensor (INA219). The different measurements are powered and collected by a microcontroller (Arduino Mega 2560) which sends the data over serial to a PC.



Figure 1: Test set-up.

Theory

The differential equation governing the steady-state behavior of a DC motor is as follows:

$$V = i_a \cdot R_a + \dot{\theta} \cdot K_m \quad (1)$$

Where V = applied voltage, i_a = motor current, R_a = total resistance of the motor, $\dot{\theta}$ = angular velocity of the rotor and K_m = motor constant.

Therefore to model the DC motor, we need the two following parameters:

- R_a - The total resistance of the motor $[\Omega]$.
- K_m - The motor constant.

Angular velocity $\dot{\theta}$ is found by taking the derivative of the rotor position θ with respect to time:

$$V_r = \dot{\theta} = \frac{d}{dt}(\theta) \quad (2)$$

The total motor resistance be found by applying the following formula to do the dataset:

$$R_a = \frac{V - V_{shunt}}{i_a} \quad (3)$$

Back-EMF ($E = \dot{\theta} \cdot K_m$) can be found with the following formula:

$$E = K_m \dot{\theta} = (V - i_a \cdot R_a - V_{shunt}) \quad (4)$$

Then to determine the motor constant K_m we can use linear regression [1]:

$$\vec{\beta} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = (X^T X)^{-1} \cdot X^T \vec{y} \quad (5)$$

$$\vec{\beta} = \begin{bmatrix} \varepsilon \\ K_m \end{bmatrix}, \quad x = \dot{\theta}, \quad y = E$$

Method

Test Procedure

Two tests were performed

- A locked rotor test.
- No load test.

The locked rotor test is performed to determine the internal resistance of the motor, when the rotor is locked, $\dot{\theta} = 0$, which means $E = 0V$, leaving only the voltage drop from the internal resistance.

Data Post-Processing

To find the angular velocity, we took the derivative of the rotor position, resulting in figure 2 for test sample 3.

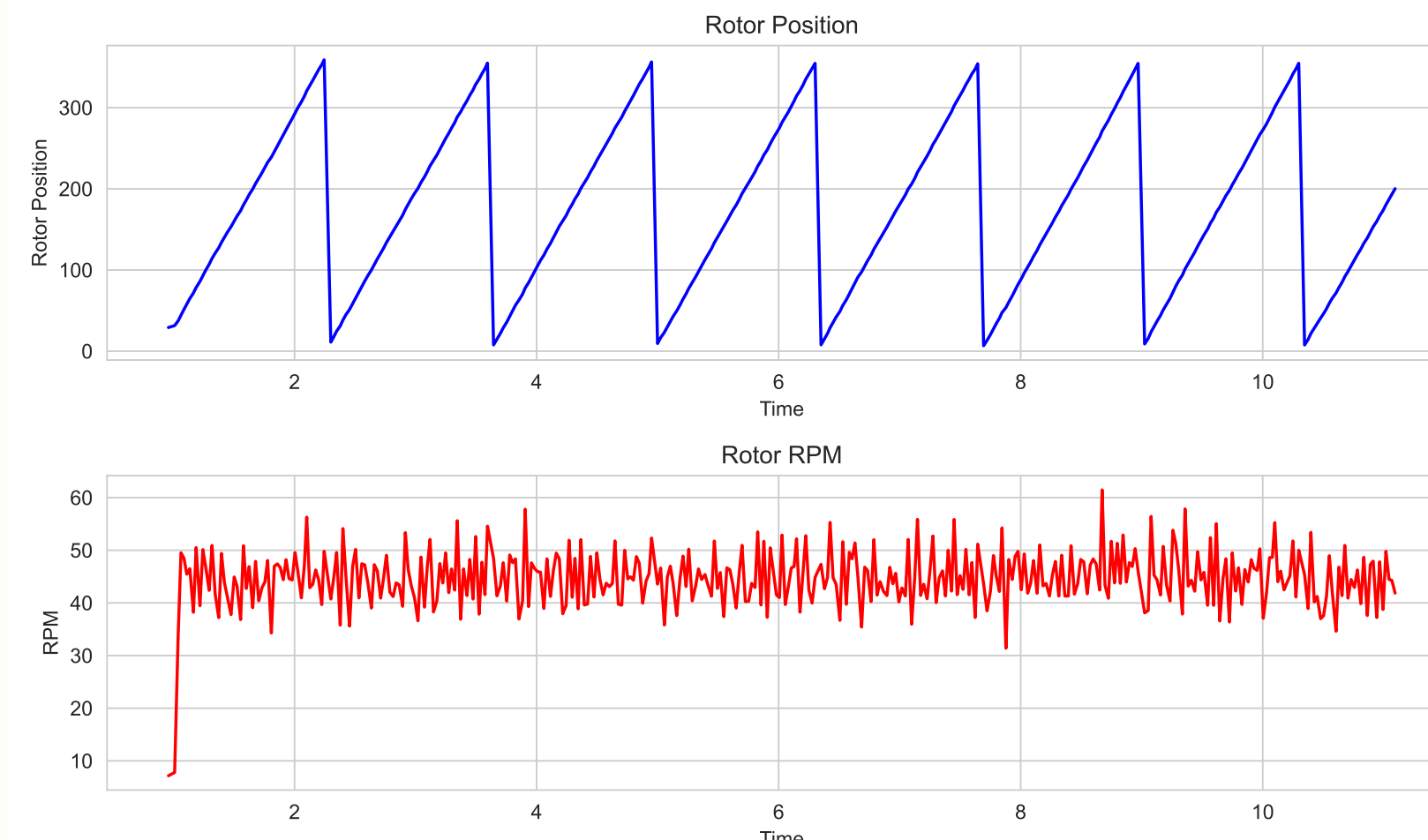


Figure 2: Rotor position (top) and estimated angular velocity (bottom).

Next, to get a realistic voltage estimate, we first needed to calculate the internal resistance of the motor using equation 3. We then applied this to equation 4 to find the back-EMF E . The results spread of data can be seen for applied voltage V in the upper plot, while the spread with the estimated back-EMF E is seen in the lower plot, in figure 3.

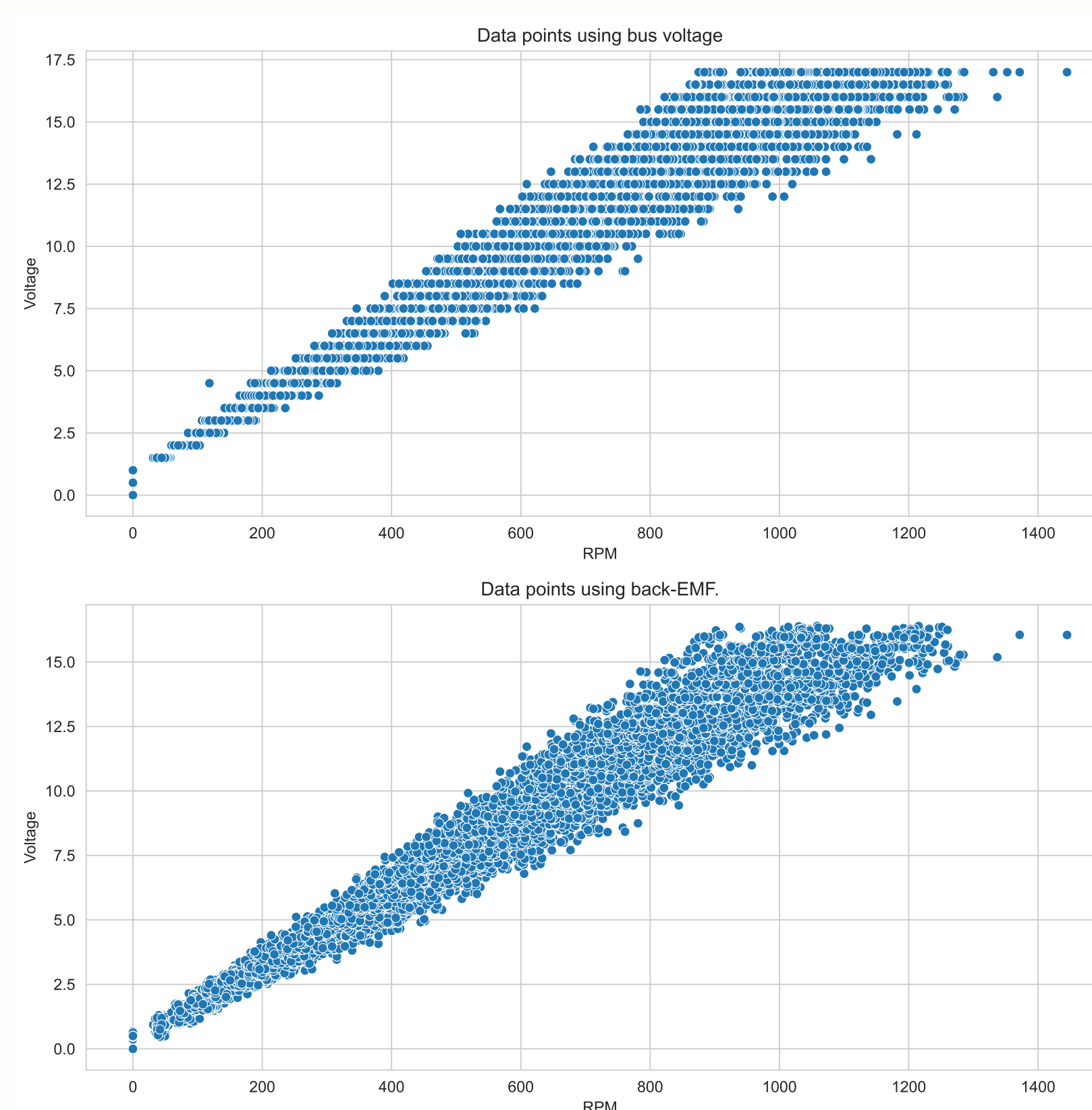


Figure 3: A comparison between applied voltage data spread (top) vs estimated back-EMF (bottom).



Scan the QR code above to access the Github repo for the full report and code.

Results

Credibility and Predictive interval at 99% confidence for linear regression line.

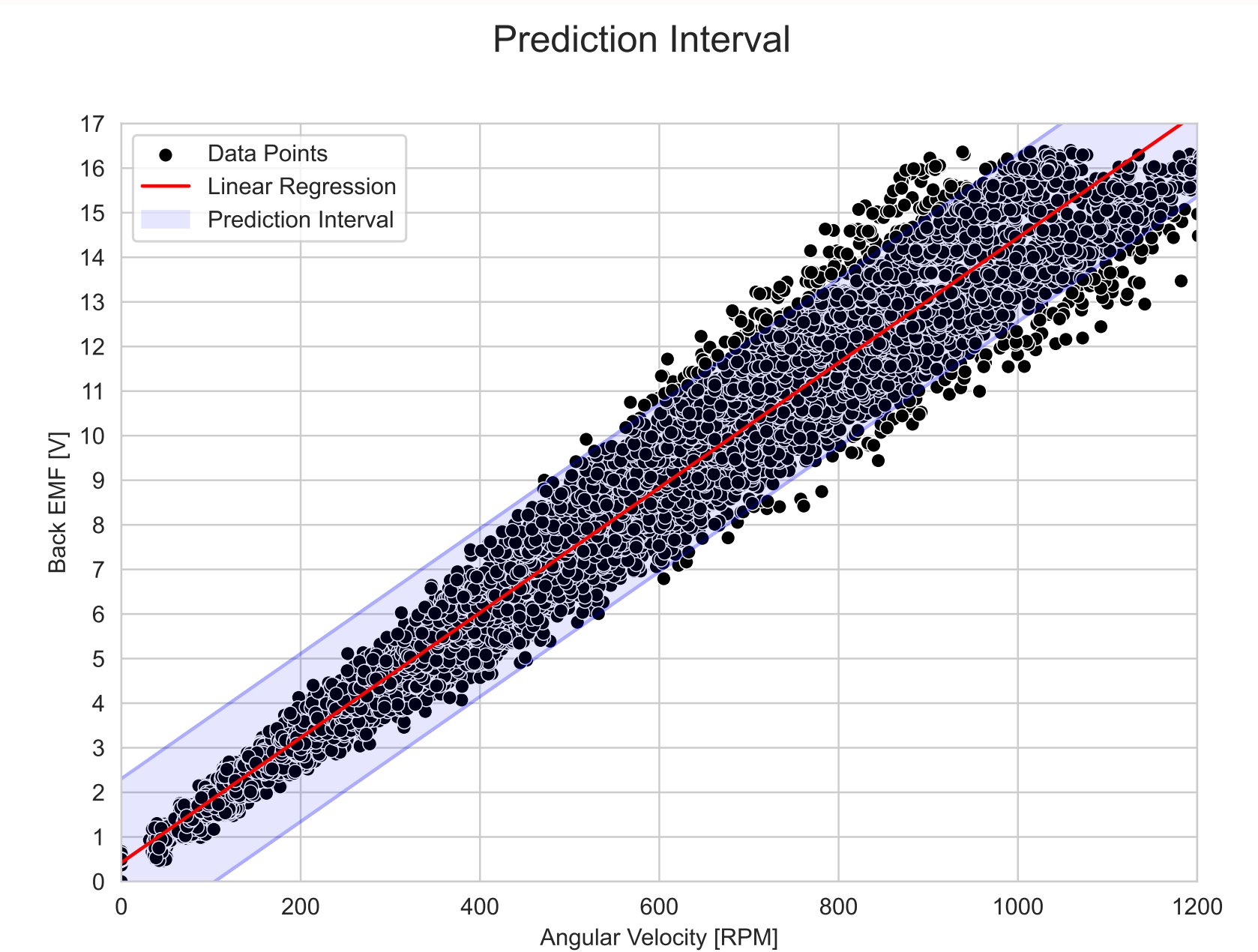


Figure 4: Prediction interval at 99% confidence.

Posterior Hyperparameter

$P1| =$

- $\bar{x} = 463.13$
- $\vec{\beta} = \begin{bmatrix} \alpha_* \\ \beta \end{bmatrix} = \begin{bmatrix} 6.90 \\ 0.014 \end{bmatrix}$
- $v_1 = 10614$
- $S_1 = 7093.64$
- $r^2 = 0.9707$

Posterior Distributions

- $\tau \sim \mathcal{U}(5307.00, 3546.8218)$
- $\alpha_* \sim t_{(6.90, 7.9344 \cdot 10^3, 10614)}$
- $\beta \sim t_{(0.014, 2.3605 \cdot 10^5, 10614)}$

Credibility and Predictive interval at 800 RPM with 99% confidence.

- $I_{0.01}(800RPM) = (11.5861V, 11.6383V)$
- $I_{0.01}^+(800RPM) = (9.7099V, 13.5145V)$

Resulting DC Motor Equation

$$V = 5.9\Omega \cdot i_a + 0.014 \cdot \dot{\theta} \quad (6)$$

Conclusion

In conclusion we designed and built a test set-up and tested a brushed DC motor from 0-15V, in 0.5V increments. Performing a locked-rotor test to estimate the internal resistance of the motor, then performing a no-load test to determine the back-EMF constant $[E]$, using statistical linear regression. We then performed inference on the model to determine the accuracy.

With a $r^2 = 0.97$ and a credibility interval at % which is hard to spot on the graph, we are happy with our results.

References

- [1] Svein Olav Nyberg. "Statistical formulas". In: Amazon Fulfillment, 2021. Chap. 14.